

How to connect Data Hub with Seven Sensor weather station



1、 Overview

1.1、 What a weather station can do?

1、 Collect ambient environment data , such as temperature, humidity, illuminance, wind speed etc.

2、 Send environment data to 3rd party control devices, and these devices can do some smart control functions.

1.2、 About Seven Sensor

Seven Sensor Solutions is located in Corum Industrial Zone in Turkey and specialized in manufacturing high-quality meteorological sensors and complete measuring & monitoring systems for solar PV plants. Seven Sensor products are used in more than 60 countries, main market is Europe.

2、 System Overview

2.1、 Datahub+Weather Station

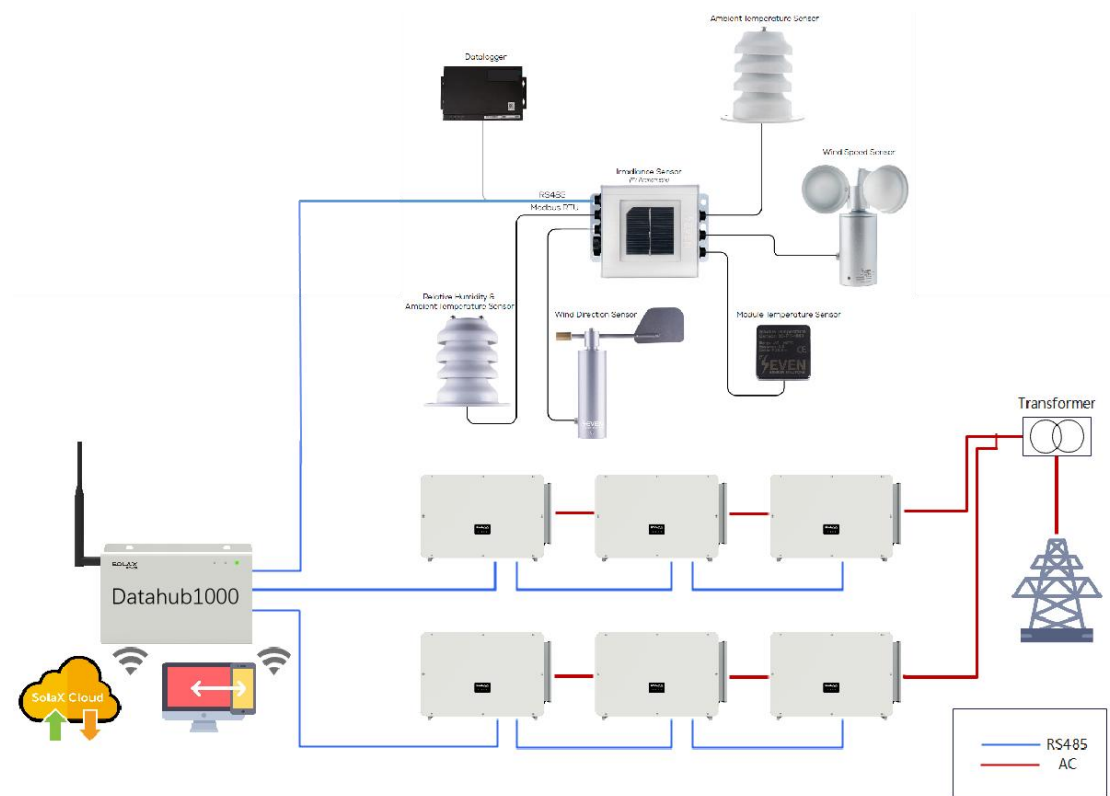
The sensor box collects environment data from the sensors connected to it. The detected data includes ambient temperature, humidity, module temperature, wind speed and wind direction etc.

The sensor box is connected to one port of datahub by RS485, and datahub can then display the weather data in the Overview page.

2.2、 Datahub Part Number

The current system datahub model is Datahub 1000, and the firmware

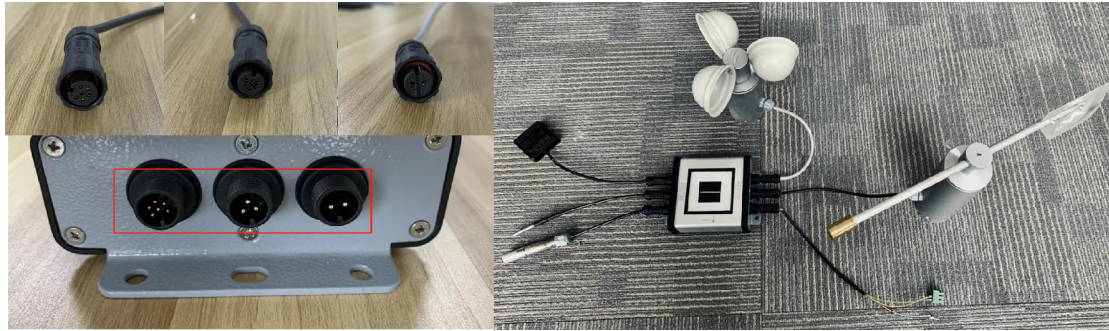
version must be V011.00 or any newer version.



3、Cable Connection

3.1、Weather Station Connection

- 1、Take SEVENSENSOR 3S-IS-5 sensor box as an example, it has 6 ports, 5 sensor ports + 1 communication port. All ports are M12 type connections with different PIN numbers, preventing wrong connections of sensor positions.
- 2、Find the correct PIN sensor, plug it to the sensor box according to the PIN number, then tighten the fastening head.
- 3、Repeat the same process to finish the connection of all the sensors.

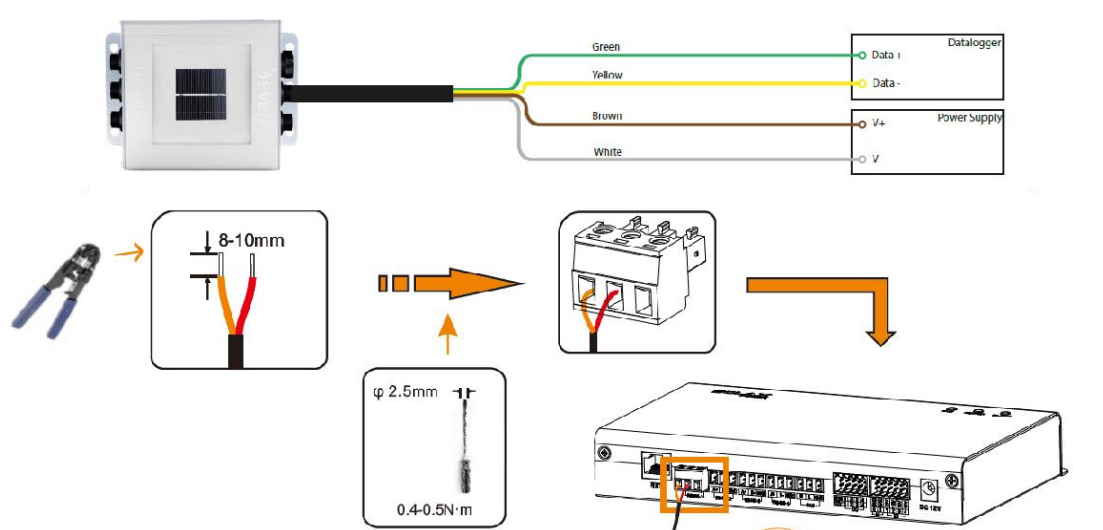


3.1、 Datahub Connection

1、 On SEVENSENSOR 3S-IS-5 PIN number 6 is the port for RS485 connection, also for power input of the sensor box, green and yellow cables are for RS485, brown and white cables are for power supply.

2、 Find the green terminal in the datahub package, remove 8-10mm insulation of green and yellow cable ends, and then insert the stripped the cables into the first 2 PINs of the green terminal respectively.

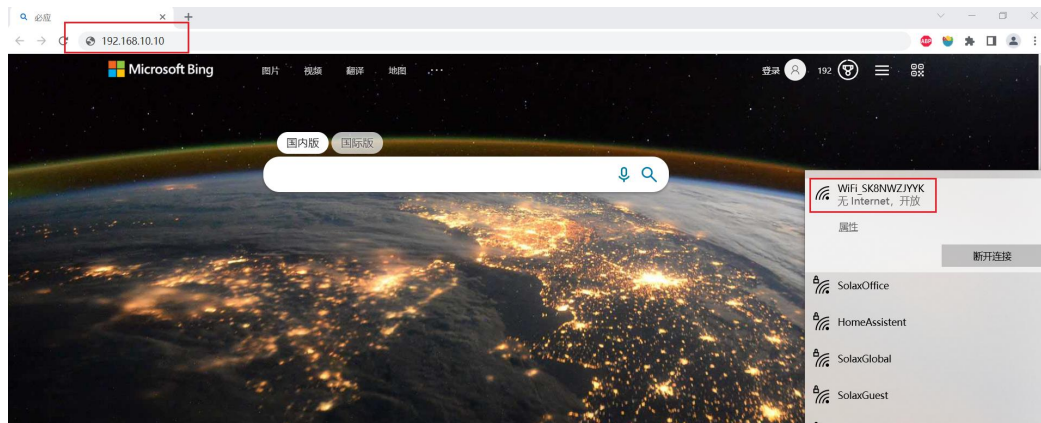
3、 Plug the terminal to one of Datahub RS485 ports.



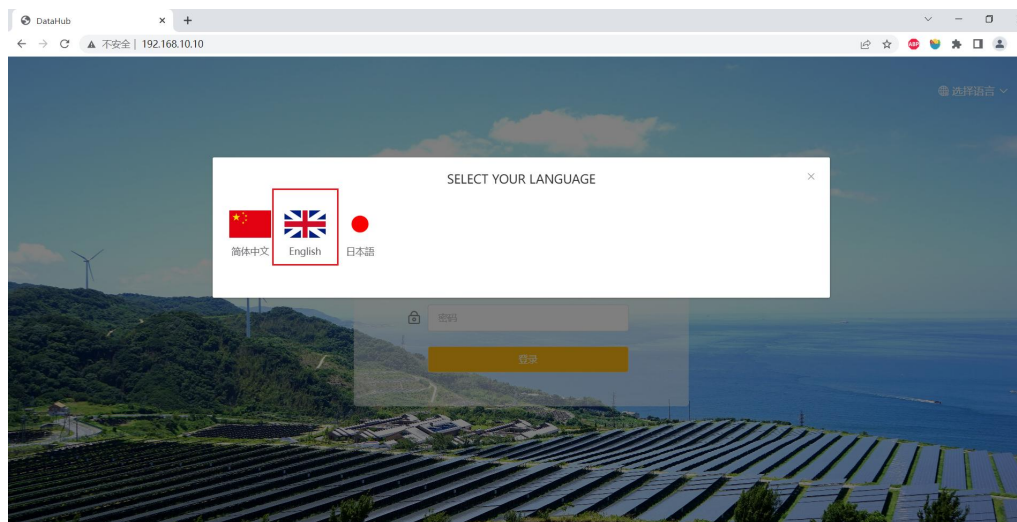
4、 Datahub System setting

1、 After the datahub is powered on, connect to WIFI, the WIFI name is

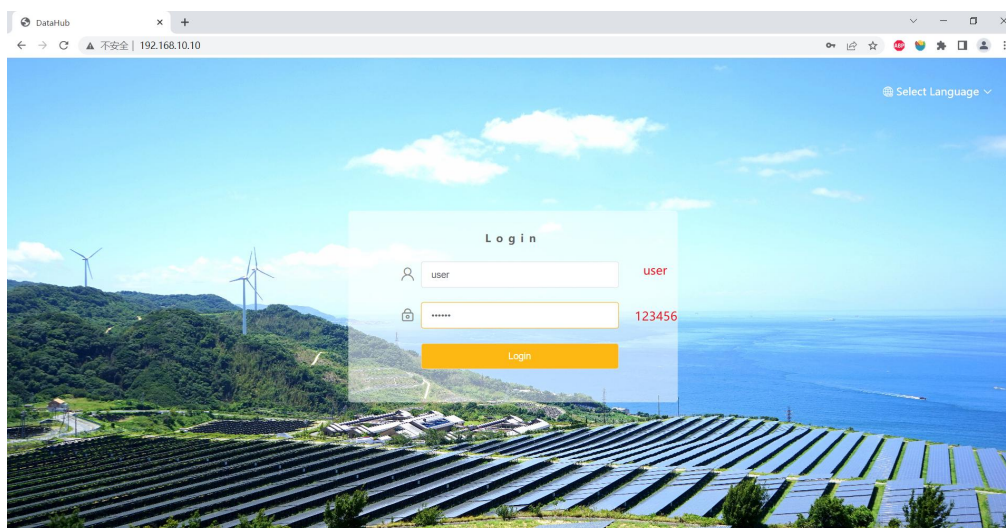
WIFI+registered, and then enter 192.168.10.10 in the browser.



2、Select Language

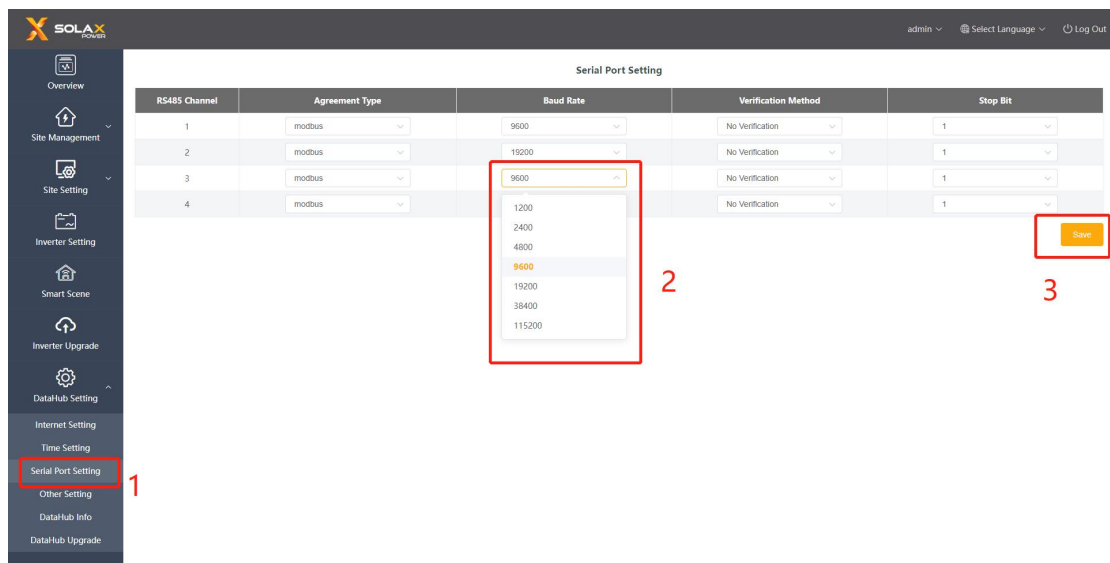


3、Enter account and password, account: user password: 123456, click login.



4、Serial port setting

- 1) In datahub web menu, find datahub setting-serial port setting
- 2) For channel 3 where the sensor box is connected, set baud rate to be 9600.
- 3) Save the settings



5、Add Device

- 1) In datahub web menu, find site management-Add device
- 2) In the corresponding channel, select device type as “environment monitor”.
- 3) Set device initial RS485 address, by default is 1 and normally no need to modify.
- 4) Set number of devices, for environment sensor, just need to input the number of sensor boxes.
- 5) Save the settings.
- 6) Once click save in the previous step, datahub will search devices on the

setting ports and display the devices found, check the name, SN, device type of the environment sensor.

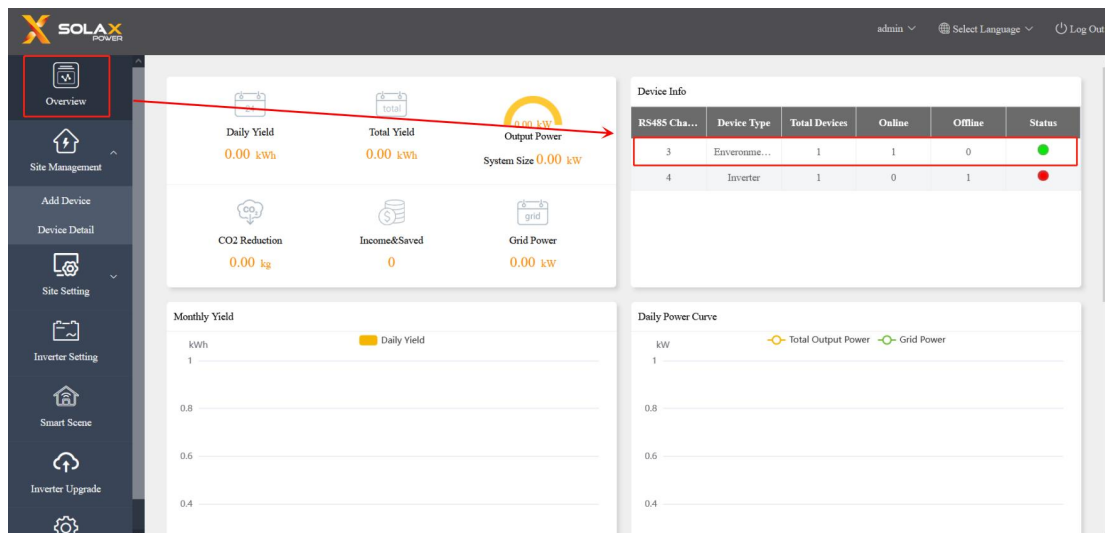
7) Click “save” again in this page to finish device add.

RS485 Channel	Device Type	Initial Address	Number of Devices
1	Meter	1	1
2	Inverter	1	1
3	Environment Monitor	1	1
4	Inverter	0	0

RS485 Channel	Device ID	SN	Alias	Device Type	Device Type
3	1	38-15-1	35-15-1	Environment Monitor	35-15
4	1	unknown	unknown	Inverter	unknown

6、Check Status

Back to the overview, the upper right corner shows that the environmental monitor is online, and green means it is online.



7、 Finally, in the overview, select the scroll bar on the right, and the chart in the lower left corner is the data of environmental monitoring. Click "<" or ">" to switch the charts of different parameters

